

Borrowed Technology

Research Design Module 3

Kormendi and Lipe (1987), *Journal of Business*

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The same question, asked twice

Macroeconomics, 1981

How much should *consumption* respond to a \$1 *income* innovation?

Hall (1978); Flavin (1981)

Accounting, 1987

How much should *price* respond to a \$1 *earnings* innovation?

Kormendi and Lipe (1987)

The second question was answered by importing the first question's structure and mathematics.

Accounting knew the sign; it did not know the size

Ball and Brown (1968) and two decades of successors established that earnings are informative: good news raises prices. A sign result.

“Whereas previous studies examined the sign of the relation between unexpected earnings and stock returns, we examine whether the magnitude of this relation is (positively) correlated with the revisions in expected future earnings derived from a univariate time-series model.”

Kormendi and Lipe (1987, p. 342)

No accounting design of the era could say how *large* the response to a dollar of earnings news should be.

The technology already existed, one field over

The rational-expectations permanent-income literature had spent a decade on the same problem:

- **Hall (1978):** consumption under rational expectations; KL note his discussion of the “close conformity” of the RE-PIH to models of firm valuation
- **Flavin (1981), Bilson (1981):** the response to an income innovation depends on the income process’s persistence, with an explicit mapping from estimated AR coefficients to the revision in permanent income
- **Sargent; Mishkin; Abel and Mishkin:** rational expectations imply nonlinear cross-equation restrictions that can be imposed and tested

The authors say exactly where the idea came from

“Our tests are derived in part from the macroeconomic literature on the rational expectations version of the permanent income hypothesis (RE-PIH) . . . Bilson (1981) and Flavin (1981) showed the correct method of calculating the effect of a current period income innovation on expected permanent income. More recently, Kormendi and LaHaye (1986) develop what they term ‘cross-regime’ tests of the RE-PIH. They use time-series data across 30 countries to test whether the size of the effect of an income innovation on consumption is related to the size of the effect of an income innovation on permanent income.”

Kormendi and Lipe (1987), footnote 5

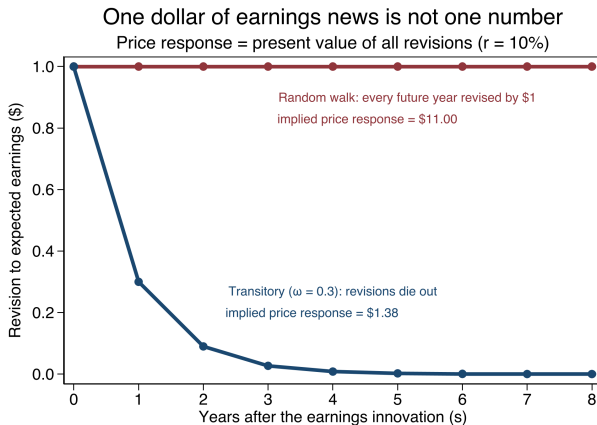
Kormendi and LaHaye (1986) is Kormendi’s *own* macro design, redeployed: income becomes earnings, consumption becomes price.

The isomorphism, object by object

Permanent-income literature	Kormendi and Lipe (1987)
income innovation	earnings innovation UX_t
revision in permanent income	PVR : PV of revisions in expected earnings
consumption response to the innovation	return response a_0
Flavin's MA-to-AR mapping	eq. (10): PVR from estimated AR coefficients
excess sensitivity of consumption	excess volatility of stock prices

Each object in the accounting design has a one-to-one macro counterpart; the estimation and testing machinery transfers with them.

Persistence determines what a dollar of news is worth



The payoff: a point prediction, not a sign prediction

The imported framework converts Miller and Rock's (1985) two-period intuition into a general prediction (their eq. 12 and eq. 10):

$$a_0 = 1 + PV R, \quad PV R = \sum_{s=1}^{\infty} \beta^s \theta_s = \frac{1}{(1 - \beta)(1 - \sum_{i=1}^N \beta^i b_i)} - 1$$

White-noise earnings: $a_0 = 1$. Random-walk earnings: $a_0 = 1 + 1/r$.

The response coefficient is pinned down by the firm's own earnings dynamics and the discount rate.

The magnitude prediction largely survives the data

145 firms, 1947–1980, joint nonlinear estimation with the cross-equation restriction imposed, on four subsets of about 50 firms (the full 582-parameter system was infeasible in 1987). The theory says $\alpha_1 = 1$:

	Subset 1	Subset 2	Subset 3	Subset 4
$\hat{\alpha}_1$	0.96	1.17	1.57	1.01
(SE)	(0.16)	(0.22)	(0.26)	(0.16)

Each estimate rejects $\alpha_1 = 0$ at the 1% level; $\alpha_1 = 1$ is rejected in only one of four subsets. The raw correlation between $\hat{a}_{0j}$ and $PV\hat{R}_j$ is 0.39 ($t = 5.0$).

The borrowed lens answers a finance debate as a by-product

Shiller (1981) and LeRoy and Porter (1981) argued aggregate prices move “too much” relative to present-value models. KL’s firm-level design asks the same question for earnings news:

$$\text{mean } \hat{a}_{0j} = 3.38 \quad \text{versus} \quad \text{mean } 1 + PV\hat{R}_j = 9.9$$

Returns respond *less* than the present-value benchmark, not more: no evidence of excess sensitivity, consistent with Kleidon (1986) and Marsh and Merton (1986). KL caveat that measurement error in the innovation biases $\hat{a}_0$ downward, so some excess sensitivity cannot be ruled out.

Good borrowing includes knowing what does not transfer

KL name three reasons the exact restriction $a_0 = 1 + PV R$ is too strong in the accounting setting:

- Earnings are not cash flows; the equivalence of the two present values is only approximate
- A univariate model understates the market's information set, so the measured innovation is an errors-in-variables proxy that biases $\hat{a}_0$ downward
- $PV R$ requires an assumed discount rate, and a wrong rate biases the benchmark

They therefore retreat, explicitly, from the exact restriction to the weaker cross-sectional test, and close with an agenda for repairing each friction.

The border was crossed twice

- Easton and Zmijewski (1989) and Collins and Kothari (1989) build directly on KL; Collins and Kothari cite Flavin (1981) themselves, so the macro original persisted into the second generation
- Fifteen years later, Vuolteenaho (2002) imports the Campbell-Shiller log-linear return decomposition at the firm level, and Callen and Segal (2004) extend it to accruals
- Vuolteenaho does *not* cite KL: the macro-finance border was crossed twice, independently, by researchers who knew the neighboring literature

We read Vuolteenaho (2002) next week.

Summary

- Kormendi and Lipe convert the information-content question from sign to magnitude by importing the present-value econometrics of the permanent-income literature; the borrowed framework yields the point prediction $a_0 = 1 + PV R$, which the data do not reject.
- The same framework provides firm-level evidence against excess sensitivity of prices to earnings news, engaging a contemporaneous finance debate as a by-product.
- Adjacent fields can supply estimation technology when problems are structurally similar; a credible transfer establishes the isomorphism and states which assumptions do not carry over.

Flavin (1981) · Kormendi and Lipe (1987) · Vuolteenaho (2002) ·
github.com/sbb151/acctg597a-research-design